

# DeepSeek V4 Pro via NVIDIA NIM

Step-by-step setup & usage guide (free tier)

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| <b>Sign-up / Console</b>                                                       |
| <code>build.nvidia.com</code>                                                  |
| <b>API Base URL</b>                                                            |
| <code>https://integrate.api.nvidia.com/v1</code>                               |
| <b>Model ID</b>                                                                |
| <code>deepseek-ai/deepseek-v4-pro-0813</code>                                  |
| <b>Model Scale</b>                                                             |
| <code>1.6T total parameters   49B active (MoE)   up to 1M token context</code> |

## Step 1 — Create your NVIDIA account

Go to [build.nvidia.com](https://build.nvidia.com) and create a free account. No credit card is required to get started.

## Step 2 — Verify your account

Complete the verification step (usually email confirmation) to unlock full access to NVIDIA NIM APIs, including free inference credits.

## Step 3 — Generate your API key

In your account settings/dashboard, generate a new API key and copy it immediately — you'll need it for every request.

## Step 4 — Find DeepSeek V4 Pro in the catalog

Search the model catalog on [build.nvidia.com](https://build.nvidia.com) for “DeepSeek V4 Pro.” The current live endpoint uses the model ID `deepseek-ai/deepseek-v4-pro-0813`.

## Step 5 — Test it in the Playground

Before writing any code, use NVIDIA's built-in browser Playground to send a few test prompts and confirm output quality and response speed.

## Step 6 — Set up the connection in code

The endpoint is OpenAI-compatible — you only change 3 things: base URL, API key, and model name. See the code block below.

```
from openai import OpenAI

client = OpenAI(
    base_url="https://integrate.api.nvidia.com/v1",
    api_key="YOUR_NVIDIA_API_KEY"
)

completion = client.chat.completions.create(
    model="deepseek-ai/deepseek-v4-pro-0813",
    messages=[{"role": "user", "content": "Hello!"}],
    temperature=1,
    top_p=0.95,
    max_tokens=16384
)

print(completion.choices[0].message.content)
```

## Step 7 — Plug it into a coding tool

Use the same base URL and API key inside Cline, Cursor, or any other OpenAI-compatible tool to bring DeepSeek V4 Pro into your coding workflow directly.

## Step 8 — Watch your usage

The free tier starts with roughly 1,000 inference credits. Monitor remaining usage on your NVIDIA dashboard so you're not caught off guard.

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Notes: DeepSeek V4 Pro is DeepSeek's flagship reasoning and coding model — a Mixture-of-Experts architecture with 1.6T total parameters and 49B active parameters, supporting up to a 1M-token context window. It is hosted directly on NVIDIA's NIM platform ([build.nvidia.com](https://build.nvidia.com)) under NVIDIA's API Trial Terms of Service. Free-tier credits and rate limits are provided for development, testing, and prototyping — production use requires NVIDIA AI Enterprise. Always confirm current terms and limits on [build.nvidia.com](https://build.nvidia.com) before relying on it in production.